

HW 1

Thursday, September 3, 2026 8:37 PM

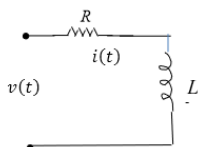
EEEE- 414: Classical Control

Homework Assignment # 1

Due date: by 11:59 PM, Thursday, September 3, 2026

Problem #1

Given the circuit shown below:



$$v(t) = i(t)R + L \frac{di(t)}{dt}$$

$$Ri(t) + L \frac{di(t)}{dt} = u(t)$$

$$L(sI(s) + i(0)) + Ri(s) = \frac{1}{s}$$

$$i(0) = 0$$

$$LsI(s) + Ri(s) = \frac{1}{s}$$

$$I(s)(Ls + R) = \frac{1}{s}$$

$$I(s) = \frac{1}{s(Ls + R)}$$

$$I(s) = \frac{\frac{1}{L}}{s(s + \frac{R}{L})}$$

$$\frac{\frac{1}{L}}{s(s + \frac{R}{L})} = \frac{A}{s} + \frac{B}{s + \frac{R}{L}}$$

$$\frac{A}{s} = \frac{\frac{1}{L}}{s(s + \frac{R}{L})} \Big|_{s=0}$$

$$A = \frac{\frac{1}{L}}{s + \frac{R}{L}} \Big|_{s=0}$$

$$A = \frac{\frac{1}{L}}{\frac{R}{L}}$$

$$A = \frac{1}{R}$$

$$\frac{B}{s + \frac{R}{L}} = \frac{\frac{1}{L}}{s(s + \frac{R}{L})} \Big|_{s = -\frac{R}{L}}$$

$$B = \frac{\frac{1}{L}}{s} \Big|_{s = -\frac{R}{L}}$$

$$B = \frac{\frac{1}{L}}{-\frac{R}{L}}$$

$$B = -\frac{1}{R}$$

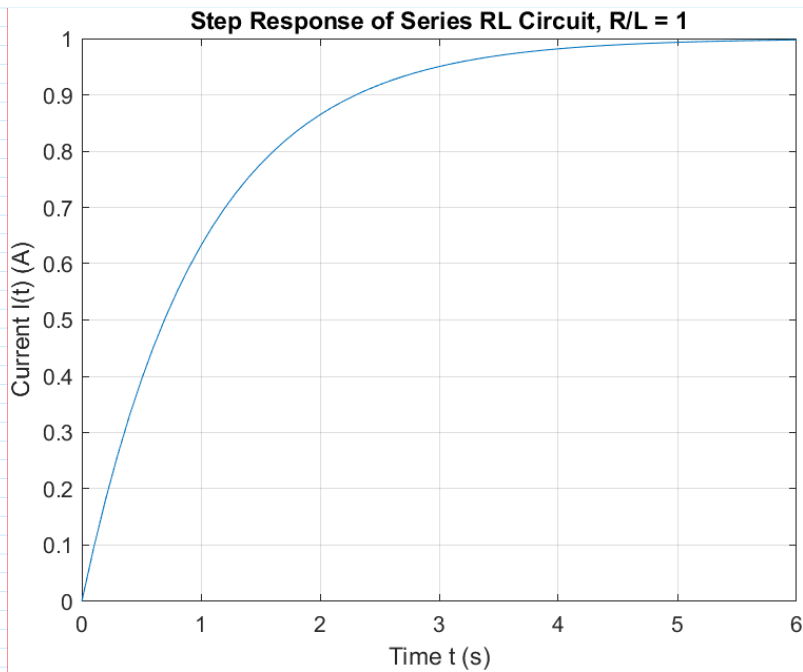
$$I(s) = \frac{1}{R} \frac{1}{s} - \frac{1}{R} \frac{1}{s + \frac{R}{L}}$$

$$I(t) = \frac{1}{R}(u(t) - e^{-\frac{R}{L}t}u(t))$$

$$I(t) = \frac{1}{R}(1 - e^{-\frac{R}{L}t})u(t)$$

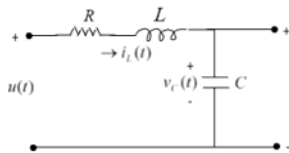
$$R/L = 1 \quad R=1 \quad L=1$$

```
Controls_Hw1_q1.m  x  +
1  R = 1;
2  L = 1;
3  t = 0:0.1:6;
4  I = (1/R) * (1 - exp(-(R/L)*t));
5
6  figure
7  plot(t, I);
8  grid on
9  xlabel('Time t (s)')
10 ylabel('Current I(t) (A)')
11 title('Step Response of Series RL Circuit, R/L = 1')
```



Problem # 2

Repeat problem #1 for the following RLC circuit. Assume $R = 1\Omega$, $L = 0.5H$, and $C = \frac{1}{8}F$



$$2A) \quad u(t) = R i_L(t) + L \frac{di_L(t)}{dt} + v_C(t)$$

$$i_L(t) = C \frac{dv_C(t)}{dt}$$

$$u(t) = \frac{1}{8} \frac{dv_C(t)}{dt} + 0.5 \frac{d}{dt} \left(\frac{1}{8} \frac{dv_C(t)}{dt} \right) + v_C(t)$$

$$u(t) = \frac{1}{8} \frac{dv_C(t)}{dt} + \frac{1}{16} \frac{d^2 v_C(t)}{dt^2} + v_C(t)$$

$$u(t) = \frac{1}{16} \frac{d^2 v_C(t)}{dt^2} + \frac{1}{8} \frac{dv_C(t)}{dt} + v_C(t)$$

$$16 u(t) = \frac{d^2 v_C(t)}{dt^2} + 2 \frac{dv_C(t)}{dt} + 16 v_C(t)$$

$$2B) \quad 16s = s^2 V_C(s) - s V_C(0) - V_C(0) + 2s V_C(s) - 2V_C(0) + 16 V_C(s)$$

$$16s = s^2 V_C(s) + 2s V_C(s) + 16 V_C(s)$$

$$16s = V_C(s) (s^2 + 2s + 16)$$

$$V_C(s) = \frac{16}{s(s^2 + 2s + 16)}$$

$$\frac{16}{s(s^2 + 2s + 16)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 16}$$

$$\frac{16}{s(s^2 + 2s + 16)} = \frac{A}{s} + \frac{S(Bs + C)}{s^2 + 2s + 16} \quad | s=0$$

$$16/16 = A + 0$$

$$A = 1$$

$$16 = A(s^2 + 2s + 16) + S(Bs + C)$$

$$16 = s^2 + 2s + 16 + Bs^2 + Cs$$

$$16 = s^2(B+1) + S(C+2) + 16$$

$$0 = s^2(B+1) + S(C+2)$$

$$1+B=0$$

$$B=-1$$

$$C+2=0$$

$$C=-2$$

$$\frac{16}{s(s^2 + 2s + 16)} = \frac{1}{s} - \frac{s+2}{s^2 + 2s + 16}$$

$$C = -2$$

$$\frac{16}{s(s^2+16)} = \frac{1}{s} - \frac{s+2}{s^2+16}$$

$$V_L(s) \rightarrow \frac{1}{s} - \frac{(s+1)+1}{(s+1)^2+16}$$

$$V_L(s) = \frac{1}{s} - \frac{s+1}{(s+1)^2+16} - \frac{1}{(s+1)^2+16}$$

$$V_L(s) = \frac{1}{s} - \frac{s+1}{(s+1)^2+16} - \frac{1}{s+1} \cdot \frac{\sqrt{16}}{(s+1)^2+16}$$

$$V_L(t) = (1 - e^{-t} \cos(\sqrt{16}t) - \frac{1}{\sqrt{16}} e^{-t} \sin(\sqrt{16}t)) u(t)$$

$$I_L(t) = \frac{1}{8} \frac{dV_L}{dt}$$

$$\frac{1}{8} \frac{d}{dt} (V_L(t)) = \frac{1}{8} \frac{d}{dt} (1 - e^{-t} \cos(\sqrt{16}t)) + \frac{1}{8} \frac{d}{dt} (-\frac{1}{\sqrt{16}} e^{-t} \sin(\sqrt{16}t)) u(t)$$

$$I_L(t) = \frac{1}{8} (e^{-t} \cos(\sqrt{16}t) + e^{-t} \sqrt{16} \sin(\sqrt{16}t)) + \frac{1}{8} (\frac{1}{\sqrt{16}} e^{-t} \sin(\sqrt{16}t) - e^{-t} \cos(\sqrt{16}t)) u(t)$$

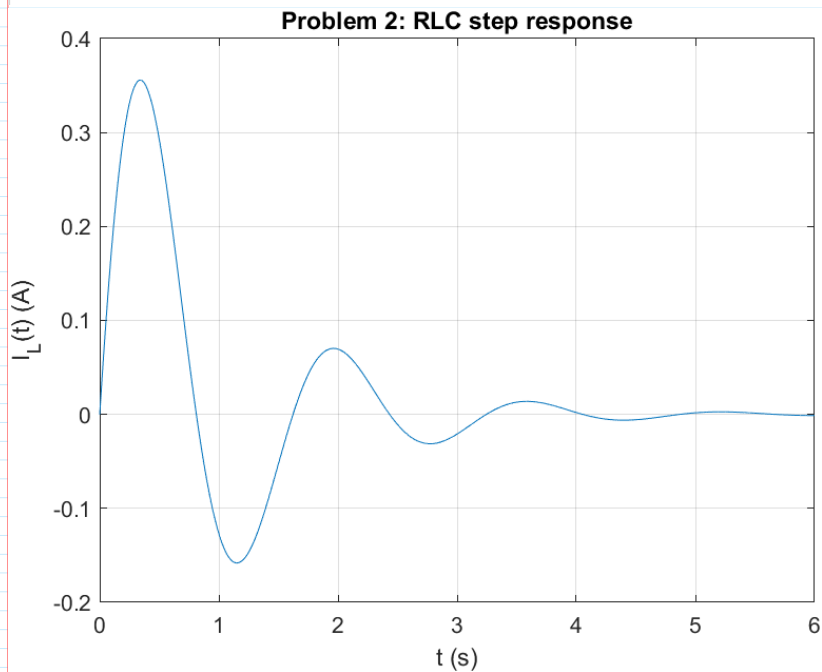
$$I_L(t) = \frac{1}{8} (e^{-t} \cos(\sqrt{16}t) + e^{-t} \sqrt{16} \sin(\sqrt{16}t) + \frac{1}{\sqrt{16}} e^{-t} \sin(\sqrt{16}t) - e^{-t} \cos(\sqrt{16}t)) u(t)$$

$$I_L(t) = \frac{1}{8} (\frac{1}{\sqrt{16}} e^{-t} \sin(\sqrt{16}t)) u(t)$$

$$I_L(t) = \frac{1}{8\sqrt{16}} e^{-t} \sin(\sqrt{16}t) u(t)$$

```
% Q2
R = 1;
L = 0.5;
C = 1/8;
t = 0:0.001:6;
I = 2/sqrt(15) * exp(-t) .* sin(sqrt(15) * t);
```

```
figure
plot(t, I);
grid on
xlabel('t (s)');
ylabel('I_L(t) (A)');
title('Problem 2: RLC step response');
```



Problem # 3

Assuming zero initial conditions, find solutions for the following differential equations.

a) $\frac{dx(t)}{dt} + 5x(t) = 2 \cos(3t)$

b) $\frac{d^2x(t)}{dt^2} + 4 \frac{dx(t)}{dt} + 2x(t) = 2 \sin(t)$

c) $\frac{d^2x(t)}{dt^2} + 6 \frac{dx(t)}{dt} + 20x(t) = 5u(t)$

3A) $sX(s) - x(0) + 5X(s) = 2 \frac{3}{s^2+9} \quad x(0)=0$

$$X(s) (s+5) = \frac{2 \cdot 3}{s^2+9}$$

$$X(s) = \frac{2s}{(s^2+9)(s+5)}$$

$$\frac{2s}{(s^2+9)(s+5)} = \frac{A}{s+5} + \frac{B+C}{s^2+9}$$

$$2s = A(s^2+9) + (Bs+C)(s+5) \quad | s = -5$$

$$-10 = A(25+9)$$

$$-10 = 34A$$

$$A = -\frac{5}{17}$$

$$2s = As^2 + 9A + Bs^2 + 5Bs + C + 5C$$

$$0 = -\frac{5}{17}s^2 - \frac{45}{17} + Bs^2 + 5Bs + C + 5C - 2s$$

$$0 = s^2(B - \frac{5}{17}) + s(5B - 2) + (C - \frac{45}{17} + 5C)$$

$$B - \frac{5}{17} = 0$$

$$B = \frac{5}{17}$$

$$5B + C - 2 = 0$$

$$5(\frac{5}{17}) + C = 2$$

$$C = \frac{9}{17}$$

$$5C - \frac{45}{17} = 0$$

$$C = \frac{9}{17}$$

$$X(s) = \frac{-\frac{5}{17}}{s+5} + \frac{\frac{5}{17}s + \frac{9}{17}}{s^2+9}$$

$$X(s) = -\frac{5}{17} \cdot \frac{1}{s+5} + \frac{1}{17} \left(\frac{5s+9}{s^2+9} \right)$$

$$X(s) = -\frac{5}{17} \cdot \frac{1}{s+5} + \frac{1}{17} \left(\frac{5s}{s^2+9} + \frac{9}{s^2+9} \right)$$

$$X(t) = -\frac{5}{17} e^{-5t} + \frac{1}{17} (5 \sin(3t) + 3 \cos(3t))$$

$$X(t) = \left(-\frac{5}{17} e^{-5t} + \frac{5}{17} \sin(3t) + \frac{3}{17} \cos(3t) \right) u(t)$$

30)

$$s^3 X(s) + 5s X(s) + X'(s) + 4s X(s) + 4 X(s) + 2 X(s) = 2 \frac{1}{s+1} \quad X(0) = 0$$

$$\frac{3}{s+1} = s^3 X(s) + 9s X(s) + 2 X(s)$$

$$\frac{3}{s+1} = X(s) (s^3 + 9s + 2)$$

$$X(s) = \frac{2}{(s+1)(s^3+9s+2)} \quad s = -1 \pm \sqrt{16-4(9)} \Rightarrow s = -2 \pm \sqrt{2}$$

$$X(s) = \frac{2}{(s+1)(s+2-\sqrt{2})(s+2+\sqrt{2})}$$

$$\frac{2}{(s+1)(s+2-\sqrt{2})(s+2+\sqrt{2})} = \frac{A}{s+2-\sqrt{2}} + \frac{B}{s+2+\sqrt{2}} + \frac{C+D}{s^2+1}$$

$$2 = A(s+2)(s^2+1) + B(s+1)(s^2+1) + (C+D)(s^2+9s+2) \quad | s = -2+\sqrt{2}$$

$$2 = A(-2+\sqrt{2}+1)(-2+\sqrt{2}+1) + 0 + 0$$

$$2 = A(2\sqrt{2})(7-4\sqrt{2})$$

$$2 = A(14\sqrt{2} - 16)$$

$$A = \frac{7\sqrt{2}+8}{34}$$

$$2 = A(s+2)(s^2+1) + B(s+1)(s^2+1) + (C+D)(s^2+9s+2) \quad | s = -2-\sqrt{2}$$

$$2 = B(-2-\sqrt{2}+1)(-2-\sqrt{2}+1) + 0 + 0$$

$$2 = B(-2\sqrt{2})(4\sqrt{2}+7)$$

$$2 = B(-16 - 14\sqrt{2})$$

$$B = -\frac{7\sqrt{2}+8}{34}$$

$$2 = A(s+2)(s^2+1) + B(s+1)(s^2+1) + (C+D)(s^2+9s+2) \quad | s = 0$$

$$2 = (C+D)(-1+9+2) + 0 + 0$$

$$2 = -C - 4C + 2C + D + 9D + 2D$$

$$2 = C(2) - 3 - 4 + D(4) + 2 - 1$$

$$2 = C(-1) + D(4) + 1$$

$$2 = -4C + C + D + 4D$$

$$2 = -4C + D \Rightarrow D = 2+4C$$

$$0 = C + 4D$$

$$0 = C + 4(2+4C)$$

$$0 = C + 8 + 16C$$

$$-8j = 19Cj$$

$$C = -8/19$$

$$Z = D - 4C$$

$$Z = D - 4(-8/19)$$

$$Z = D + 32/19$$

$$D = 2/19$$

$$X(s) = \frac{10s - 8}{s^2 - \sqrt{2}} - \frac{14s - \frac{8}{\sqrt{2}}}{s^2 + \sqrt{2}} + \frac{-8/19s + 2/19}{s^2 + 1}$$

$$X(t) = \left(7\sqrt{2} - \frac{8}{\sqrt{2}} e^{-\frac{\sqrt{2}}{2}t} - 7\sqrt{2} - \frac{8}{\sqrt{2}} e^{-\frac{\sqrt{2}}{2}t} - \frac{8}{\sqrt{2}} (\cos(t) + \frac{2}{\sqrt{2}} \sin(t)) \right) u(t)$$

3c)

$$s^2 X(s) - s X(0) - X'(0) + 6s X(s) - 6 X(0) + 20 X(s) = 5/3$$

$$5/3 = s^2 X(s) + 6s X(s) + 20 X(s)$$

$$5/3 = X(s) (s^2 + 6s + 20)$$

$$X(s) = \frac{5/3}{s^2 + 6s + 20}$$

$$\frac{5}{s^2 + 6s + 20} = \frac{A}{s} + \frac{Bs + C}{s^2 + 6s + 20}$$

$$5 = A(s^2 + 6s + 20) + (Bs + C)s$$

$$5 = A(20)$$

$$A = 1/4$$

$$5 = A(s^2 + 6s + 20) + (Bs + C)s$$

$$5 = \frac{1}{4}s^2 + \frac{6}{4}s + \frac{5}{2} + \frac{Bs^2}{4} + \frac{Cs}{4}$$

$$0 = s^2 \left(\frac{1}{4} + B \right) + s \left(\frac{6}{4} + C \right)$$

$$\frac{1}{4} + B = 0$$

$$B = -1/4$$

$$\frac{6}{4} + C = 0$$

$$C = -3/2$$

$$X(s) = \frac{1/4}{s} - \frac{\frac{1}{4}s + \frac{3}{2}}{s^2 + 6s + 20}$$

$$X(s) = \frac{1/4}{s} - \frac{1}{4} \cdot \frac{s+3}{(s+4)(s+2)} = \frac{1/4}{s} - \frac{1}{4} \cdot \frac{s+3}{(s+4)(s+2)}$$

$$X(t) = \left(\frac{1}{4} - \frac{1}{4} e^{-2t} (\cos(\sqrt{2}t) - \frac{2}{\sqrt{2}} e^{-3t} \sin(\sqrt{2}t)) \right) u(t)$$

Problem # 4

Solve the following differential equations:

a) $\frac{d^2 x(t)}{dt^2} + 2 \frac{dx(t)}{dt} + 2x(t) = \sin(2t)$

$$x(0) = 2$$

$$\frac{dx(t)}{dt} (t=0) = -3$$

b) $\frac{d^2 x(t)}{dt^2} + 2 \frac{dx(t)}{dt} + x(t) = 5e^{-2t} + t$

$$x(0) = 2$$

$$\frac{dx(t)}{dt} (t=0) = 1$$

c) $\frac{d^2 x(t)}{dt^2} + 4x(t) = t^2$

$$x(0) = 1$$

$$\frac{dx(t)}{dt} (t=0) = 2$$

4a)

$$s^2 X(s) - s X(0) - X'(0) + 2s X(s) - 2 X'(0) + 2 X(s) = \frac{2}{s^2 + 4}$$

$$s^2 X(s) - 2s + 3 + 2s X(s) - 4 + 2 X(s) = \frac{2}{s^2 + 4}$$

$$X(s) (s^2 + 2s + 2) - 2s - 1 = \frac{2}{s^2 + 4}$$

$$X(s) (s^2 + 2s + 2) = \frac{2}{s^2 + 4} + 2s + 1$$

$$X(s) = \frac{2 + (s^2 + 1)(2s + 1)}{(s^2 + 4)(s^2 + 2s + 2)}$$

$$X(s) = \frac{2s^3 + s^2 + 2s + 1}{(s^2 + 4)(s^2 + 2s + 2)}$$

$$\frac{2s^3 + 15s^2 + 8s + 6}{(s^2 + 4)(s^2 + 23s + 12)} = \frac{As + B}{s^2 + 4} + \frac{Cs + D}{s^2 + 23s + 12}$$

$$2s^3 + 15s^2 + 8s + 6 = (As + B)(s^2 + 23s + 12) + (Cs + D)(s^2 + 4)$$

$$2s^3 + 15s^2 + 8s + 6 = As^3 + 23As^2 + 24As + 12Bs + 12B + Cs^3 + 4Cs + Ds^2 + 4D$$

$$2 = A + C$$

$$15 = 23A + B + D$$

$$8 = 24A + 12B + 4C$$

$$6 = 12B + 4D$$

$$A = -1/5$$

$$B = -1/5$$

$$C = 11/5$$

$$D = 8/5$$

$$\chi(s) = -\frac{1}{5} \frac{s-1}{s^2+4} + \frac{11}{5} \frac{s+1}{s^2+23s+12}$$

$$\chi(s) = -\frac{1}{5} \frac{s-1}{s^2+4} - \frac{1}{10} \frac{2}{s^2+4} + \frac{11}{5} \frac{(s+1)}{(s+1)^2+1} - \frac{8}{5} \frac{s}{(s+1)^2+1}$$

$$\chi(t) = \left(\frac{1}{5} e^{-t} \cos(t) - \frac{3}{5} e^{-t} \sin(t) - \frac{1}{10} \cos(2t) - \frac{1}{10} \sin(2t) \right) u(t)$$

4b)

$$s^2 \chi(s) - s \chi(0) - \chi'(0) + 2s \chi(s) - 2 \chi(0) + \chi(s) = \frac{2}{s+2} + \frac{1}{s^2}$$

$$s^2 \chi(s) - 2s - 1 + 2s \chi(s) - 4 + \chi(s) = \frac{2}{s+2} + \frac{1}{s^2}$$

$$\chi(s) (s^2 + 2s + 1) - 2s - 5 = \frac{2}{s+2} + \frac{1}{s^2}$$

$$\chi(s) (s^2 + 2s + 1) = \frac{s^2 + 5s + 2}{s^2 (s+2)}$$

$$\chi(s) (s^2 + 2s + 1) = \frac{s^2 + 5s + 2 + (2s+1)(s^2+2s)}{s^2 (s+2)}$$

$$\chi(s) = \frac{2s^4 + 9s^3 + 15s^2 + 5s + 2}{s^2 (s+2) (s^2 + 2s + 1)}$$

$$\frac{2s^4 + 9s^3 + 15s^2 + 5s + 2}{s^2 (s+2) (s+1)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{s+1} + \frac{E}{(s+1)^2}$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = E(s^2)(s+1) + D(s+1)(s+2) + C(s^2)(s+1)^2 + B(s+2)(s+1)^2 + A(s+2)(s+1)^2$$

$$2s - 9 + 15s - 12 = E$$

$$E = 9$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = 9(s+2)(s+1)^2 + 0 + 0 + 0 + 0 \quad | s=0$$

$$2 = 0(E)$$

$$0 = 1$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = C(s^2)(s+1)^2 + 0 + 0 + 0 + 0 \quad | s=-2$$

$$32 - 72 + 60 - 2 + 2 = C(4)$$

$$1/4 = 20$$

$$C = 5$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = A(s)(s+2) + B(s+2)(s+1)^2 + C(s^2)(s+1)^2 + D(s+1)(s+2) + E(s^2)(s+1)^2$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = A(s^4 + 4s^3 + 6s^2 + 4s + 2) + B(s^3 + 6s^2 + 9s + 2) + C(s^4 + 4s^3 + 6s^2 + 4s + 2) + D(s^3 + 3s^2 + 2s) + E(s^4 + 4s^3 + 6s^2 + 4s + 2)$$

$$2s^4 + 9s^3 + 15s^2 + 5s + 2 = As^4 + 4As^3 + 6As^2 + 4As + 2A + Bs^3 + 6Bs^2 + 9Bs + 2B + Cs^4 + 4Cs^3 + 6Cs^2 + 4Cs + 2C + Ds^3 + 3Ds^2 + 2Ds + Es^4 + 4Es^3 + 6Es^2 + 4Es + 2E$$

$$2 = A + 2C + 2E$$

$$9 = 4A + 11B + 3D + 9E$$

$$15 = 5A + 4C + 5D + 15E$$

$$1 = 2A + 5$$

$$2 = 2E$$

$$A = -2$$

$$D = -1$$

$$\chi(s) = -\frac{2}{s} + \frac{1}{s^2} + \frac{5}{s+2} - \frac{1}{s+1} + \frac{9}{(s+1)^2}$$

$$\chi(s) = (-2 + 1 + 5e^{-2t} - e^{-t} + 9te^{-t}) u(t)$$

4c)

$$s^2 \chi(s) - s \chi(0) - \chi'(0) + 4 \chi(s) = \frac{2}{s^3}$$

$$s^2 \chi(s) - s - 2 + 4 \chi(s) = \frac{2}{s^3}$$

$$\chi(s) (s^2 + 4) - s - 2 = \frac{2}{s^3}$$

$$\chi(s) (s^2 + 4) = \frac{2 + s^3 + 2s}{s^3}$$

$$s^2 X(s) - s - 2 + 4X(s) = \frac{2}{s^3}$$

$$X(s) (s^2 + 4) \cdot s - 2 = \frac{2}{s^3}$$

$$X(s) (s^2 + 4) = \frac{2 + s^4 + 2s^3}{s^3}$$

$$X(s) = \frac{s^4 + 2s^3 + 2}{s^3(s^2 + 4)}$$

$$\frac{s^4 + 2s^3 + 2}{s^3(s^2 + 4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{Ds + E}{s^2 + 4}$$

$$s^4 + 2s^3 + 2 = C(s^2 + 4) + 0 + 0 + 0 + 0 + 0$$

$$2 = C(4)$$

$$C = \frac{1}{2}$$

$$s^4 + 2s^3 + 2 = A(s^4/s^2 + 4) + B(s^3/s^2 + 4) + C(s^2 + 4) + (Ds + E)(s^2)$$

$$s^4 + 2s^3 + 2 = As^4 + 4As^2 + Bs^3 + 4Bs + Cs^2 + 4C + Ds^3 + Es^2 + 2Ds + 2E$$

$$s^4 + 2s^3 + 2 = As^4 + 4As^2 + Bs^3 + 4Bs + Cs^2 + 4C + Ds^3 + Es^2 + 2Ds + 2E$$

$$1 = A + 0$$

$$2 = 0 + 4C$$

$$0 = 4A + \frac{1}{2}$$

$$0 = 4B$$

$$2 = 2E$$

$$0 = 0$$

$$C = 2$$

$$A = -\frac{1}{8}$$

$$D = \frac{9}{8}$$

$$X(s) = -\frac{1}{8} + \frac{1/2}{s^3} + \frac{9/8s}{s^2 + 4} + \frac{2}{s^2 + 4}$$

$$X(t) = \left[-\frac{1}{8} + \frac{1}{4}t^2 + \frac{9}{8}\cos(2t) + \frac{1}{4}\sin(2t) \right] u(t)$$

Problem # 5

A system is described by the following differential equation:

$$\frac{d^3 y(t)}{dt^3} + 3 \frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + y(t) = \frac{d^3 x(t)}{dt^3} + 4 \frac{d^2 x(t)}{dt^2} + 6 \frac{dx(t)}{dt} + 8x(t)$$

Find the expression for the transfer function of the systems $H(s) = \frac{Y(s)}{X(s)}$

$$s^3 Y(s) - s^2 y(0) - s \dot{y}(0) - \ddot{y}(0) + 3s^2 Y(s) - 3s \dot{y}(0) - 3 \ddot{y}(0) + 5s Y(s) - 5 \dot{y}(0) + Y(s) = s^3 X(s) - s^2 x(0) - s \dot{x}(0) - \ddot{x}(0) + 4s^2 X(s) - 4s \dot{x}(0) - 4 \ddot{x}(0) + 6s X(s) - 6 \dot{x}(0) + 8X(s)$$

$$\dot{y}(0) = 0 \quad \ddot{y}(0) = 0 \quad \ddot{y}(0) = 0 \quad \dot{x}(0) = 0 \quad \ddot{x}(0) = 0 \quad \ddot{x}(0) = 0$$

$$s^3 Y(s) + 3s^2 Y(s) + 5s Y(s) + Y(s) = s^3 X(s) + 4s^2 X(s) + 6s X(s) + 8X(s)$$

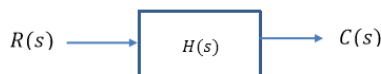
$$Y(s) (s^3 + 3s^2 + 5s + 1) = X(s) (s^3 + 4s^2 + 6s + 8)$$

$$H(s) = \frac{Y(s)}{X(s)}$$

$$H(s) = \frac{s^3 + 4s^2 + 6s + 8}{s^3 + 3s^2 + 5s + 1}$$

Problem # 6

Write the differential equation for the system shown below:



$$\frac{C(s)}{R(s)} = \frac{s^5 + 2s^4 + 4s^3 + 4s^2 + 4}{s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5}$$

$$C(s) (s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5) = R(s) (s^5 + 2s^4 + 4s^3 + 4s^2 + 4)$$

$$\frac{d^6 C(t)}{dt^6} + 7 \frac{d^5 C(t)}{dt^5} + 3 \frac{d^4 C(t)}{dt^4} + 2 \frac{d^3 C(t)}{dt^3} + \frac{d^2 C(t)}{dt^2} + 5 C(t) = \frac{d^5 R(t)}{dt^5} + 2 \frac{d^4 R(t)}{dt^4} + 4 \frac{d^3 R(t)}{dt^3} + 4 \frac{d^2 R(t)}{dt^2} + 4 R(t)$$

Problem # 7

Use MATLAB to generate the partial fraction expansion of the following transfer function.

$$F(s) = \frac{10^4(s+5)(s+70)}{s(s+45)(s+55)(s^2+7s+110)(s^2+6s+95)}$$

```
% Q7
syms s
F = 1e4*(s+5)*(s+70) / (s*(s+45)*(s+55)*(s^2+7*s+110)*(s^2+6*s+95));
partfrac(F, s)
```

>> Controls_Hw1

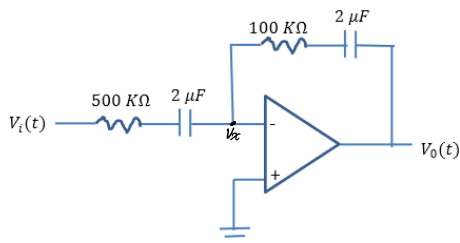
```
ans =

200/(30303*(s + 45)) - 20/(11253*(s + 55)) + 2800/(20691*s) + ((481820*s)/253253 + 1237640/253253)/(s^2 + 7*s + 110) - ((3071600*s)/1503717 + 5452600/1503717)/(s^2 + 6*s + 95)
```

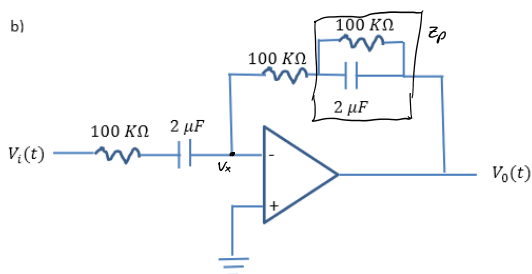
Problem # 8

Find the transfer function, $G(s) = \frac{V_0(s)}{V_i(s)}$ for each operational amplifier circuit shown below.

a)



b)



8a)

$$Z_c = \frac{1}{sC} = \frac{1}{s \cdot 2 \times 10^{-6}}$$

$$Z_c = \frac{500000}{s}$$

$$I_1 \sim \frac{V_i - V_c}{Z_c}$$

$$I_c \sim \frac{V_c - V_o}{Z_c}$$

$$I_o = 0$$

$$I_1 = I_2$$

$$\frac{V_i - V_c}{Z_c} = \frac{V_c - V_o}{Z_c}$$

$$V_c = 0$$

$$\frac{V_i}{Z_c} = -\frac{V_o}{Z_c}$$

$$\frac{V_o}{V_i} = -\frac{Z_c}{Z_f}$$

$$Z_c = 500000 + \frac{500000}{s}$$

$$Z_f = \frac{100000(s+1)}{s}$$

$$Z_f = \frac{50000}{s} + 100000$$

$$Z_f = \frac{100000(s+5)}{s}$$

$$G(s) = -\frac{Z_c}{Z_f}$$

$$G(s) = -\frac{100000(s+1)}{50000(s+5)}$$

$$G(s) = -\frac{100000(s+1)}{500000(s+5)}$$

$$G(s) = -\frac{s+1}{5(s+5)}$$

8b)

$$Z_c = 100000 + \frac{500000}{s}$$

$$Z_c = \frac{100000(s+5)}{s}$$

$$Z_f = \frac{100000}{s} + 100000$$

$$Z_f = \frac{50000}{s} + 100000$$

$$Z_f \sim \frac{500000}{s+5}$$

$$Z_f = Z_R + Z_L$$

$$Z_f \sim 100000 + \frac{500000}{s+5}$$

$$Z_f = \frac{100000(s+5) + 500000}{s+5}$$

$$Z_f = \frac{100000(5s+1) + 5}{s+5}$$

$$Z_f = \frac{100000(5s+1)}{s+5}$$

$$G(s) = - \frac{Z_f}{Z_L}$$

$$G(s) = - \frac{100000(5s+1)/s+5}{100000(5s+1)/s}$$

$$G(s) = - \frac{(s+5)/s+5}{(5s+1)/s}$$

$$G(s) = - \frac{s+5}{5s+1} \cdot \frac{s}{s+5}$$

$$G(s) = - \frac{s(s+5)}{(s+5)^2}$$

```
% Q1
R = 1;
L = 1;
t = 0:0.1:6;
I = (1/R) * (1 - exp(-(R/L)*t));
figure
plot(t, I);
grid on
xlabel('Time t (s)')
ylabel('Current I(t) (A)')
title('Step Response of Series RL Circuit, R/L = 1')

% Q2
R = 1;
L = 0.5;
C = 1/8;
t = 0:0.001:6;
I = 2/sqrt(15) * exp(-t) .* sin(sqrt(15) * t);
figure
plot(t, I);
grid on
xlabel('t (s)');
ylabel('I_L(t) (A)');
title('Problem 2: RLC step response');

% Q7
syms s
F = 1e4*(s+5)*(s+70) / (s*(s+45)*(s+55)*(s^2+7*s+110)*(s^2+6*s+95));
partfrac(F, s)
```